

Reliable Numerical Controllability of Parabolic PDEs

Publication-Oriented Research Protocol — Revised Edition, Iteration 2

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1 Preface to the revised edition

Iteration 1 revised the original dossier after an independent review that (a) directly verified the load-bearing citations rather than trusting the bibliography at face value, (b) searched the current literature (2024-2026) for preconditioning work the original cross-reference pass missed, and (c) stress-tested whether the experiment protocol and its numerical thresholds were derived or merely asserted. That pass made three changes: it demoted the preconditioning contribution (now labeled C6) from a headline claim to a conditional one; replaced the one-line theorem placeholder with an actual derivation sketch; and inserted a pilot phase before the full 165-configuration protocol, with the acceptance thresholds flagged as provisional pending that pilot.

Iteration 2 (this edition) corrects an over-correction in iteration 1: in the process of tightening the preconditioning discussion, iteration 1 discarded the original dossier’s most substantial piece of cross-referencing — the full “Seed-Based Cross-Reference Expansion” appendix (Priority A/B/C, roughly 35 papers) — and replaced it with six bullet points on preconditioning alone. That appendix is restored in full below, merged with the 2024-2026 additions rather than kept separate, and annotated against six explicit contribution labels (C1-C6) so a reader can see at a glance which literature bears on which claim. Restoring it surfaced a second correction: **item A12 in the restored appendix (Boyer & Hernandez-Santamaria, 2026) already proposes matching a relaxation scale to numerical accuracy**, which narrows the mesh-dependent regularization-law claim (C5) in the same way the 2024-2026 cluster narrowed the preconditioning claim (C6) in iteration 1. That correction is threaded through the novelty ranking, the contribution-claims section, and the final assessment below.

Everything else that held up under both review passes — the staged block design, the error-decomposition philosophy, the four-month plan, the decision tree — is retained with light edits.

2 Executive verdict on the three questions

2.1 Can this be published?

Yes, the research *direction* is publishable; no, the current *protocol*, by itself, is still not a publishable paper, because no theorem, estimator, or preconditioner has actually been produced yet — only a plan to look for one. The central publication question remains:

**How accurately, stably, and efficiently can a discrete null control be computed
as $(h, k, \varepsilon, \tau) \rightarrow 0$?**

where h is the spatial mesh size, k the time step, ε a regularization parameter, and τ an algebraic/optimization stopping tolerance. This question is not exhausted by the existing controllability literature, but after review it is *closer to exhausted on the preconditioning axis* than the original dossier assumed (see the restored Priority B appendix below). The paper must contribute at least one of:

1. a new theorem or sharp bound for the error/conditioning of a canonical controllability discretization;
2. a solver-aware stopping rule that provably or demonstrably avoids oversolving relative to the discretization error, **specific to the controllability/Gramian structure** rather than a generic tracking-type cost;
3. a preconditioner or scaling yielding mesh-independent behavior **that improves on, or is mechanistically distinct from, the current 2024-2026 preconditioning literature** (not merely a re-demonstration);
4. a rigorous a posteriori decomposition of control error into spatial, temporal, regularization, and algebraic components;
5. a quantitative comparison of HUM and primal/least-squares formulations at equal achieved accuracy, with a mathematical explanation of the observed spectral differences.

Ranked by novelty risk after review: **(2) and (4) are now the safest bets, (5) is a solid second paper, (1) is valuable but the hardest to actually prove, and (3) alone is the riskiest** — it is the crowded item.

2.2 Is it clear how and how many experiments are needed?

Mostly, with one correction: the protocol previously moved directly from “15 verification configurations” to a 123-165 configuration full study. A **pilot phase** is now inserted between them (Step 0, below) to check, on a small slice, whether the asymptotic regime and the proposed thresholds are even reachable before the full budget is spent. The block structure (V, C, S, R, P) and the 123-configuration core / 165-configuration robustness-extended study are otherwise unchanged, because review found the staged design (versus a full factorial) to be sound.

2.3 Does it reference numerical-analysis papers that use the same methodological logic?

Yes, and the core precedent set (Meidner-Vexler, Ern-Vohralik, Guttel-Pearson, Rees-Dollar-Wathen, Boyer-Hubert-Le Rousseau) was independently verified and holds up: these are real, on-topic papers. However, the original cross-reference exercise had a **recency blind spot** on exactly the axis (preconditioning) where the project claims some of its novelty. The restored and extended

appendix below adds the papers that were missing, and the effect is to narrow, not widen, where a first paper should place its bet.

3 How the research question was refined across the project

(Unchanged from the original dossier — retained for context.)

The project began from an existing 2024 master’s thesis on numerical and constrained controllability of the heat equation, which already implemented 1-D HUM, penalized HUM, conjugate gradients, implicit Euler, internal control, and boundary control, and contained the empirical clue that unpenalized HUM becomes dramatically more expensive under refinement while penalization changes the numerical behavior substantially.

The subsequent literature discussion eliminated several weak research claims in turn: showing that a control can be computed at all; showing that it converges; observing ill-conditioning (already reported by Fernandez-Cara and Munch for a primal formulation); using penalized HUM alone (already Boyer’s framework); and jumping straight to Burgers before the linear error budget is understood. The refined question is:

Given a target accuracy in the control and terminal state, determine how fine the PDE discretization must be, how regularization should scale, how accurately the discrete optimization problem must be solved, and whether preconditioning can make the computational effort predictable under refinement.

4 Mathematical baseline

4.1 Controlled heat equation

$$\partial_t y - \nu \partial_{xx} y = \mathbf{1}_\omega u \quad \text{in } (0, 1) \times (0, T), \quad y(0, t) = y(1, t) = 0, \quad y(x, 0) = y_0(x),$$

with $\nu > 0$, a nonempty control region $\omega \subset (0, 1)$, and null-control target $y(\cdot, T) = 0$. In minimal-norm HUM the discrete system takes the form $A_{h,k} p_{h,k} = b_{h,k}$, where $A_{h,k}$ is the discrete controllability Gramian/Hessian. Its condition number $\kappa_2(A_{h,k}) = \lambda_{\max}/\lambda_{\min}$ governs worst-case sensitivity and enters the classical CG estimate

$$\frac{\|e_m\|_A}{\|e_0\|_A} \leq 2 \left(\frac{\sqrt{\kappa_2(A)} - 1}{\sqrt{\kappa_2(A)} + 1} \right)^m.$$

4.2 Penalization and the bias-conditioning tradeoff

$$J_\varepsilon(u) = \frac{1}{2} \|u\|_{L^2(\omega \times (0, T))}^2 + \frac{1}{2\varepsilon} \|y(T)\|_{L^2(0, 1)}^2, \quad \varepsilon > 0.$$

Small ε enforces smaller terminal residual but may worsen conditioning; large ε stabilizes but biases the control. The question is not whether $\varepsilon \rightarrow 0$, but whether a mesh-dependent choice $\varepsilon = \varepsilon(h, k)$ can balance regularization bias against discretization and algebraic error.

4.3 Error budget

$$\|u - \tilde{u}_{h,k}^{\varepsilon, \tau}\| \leq \underbrace{\|u - u_{h,k}\|}_{\text{space/time discretization}} + \underbrace{\|u_{h,k} - u_{h,k}^\varepsilon\|}_{\text{regularization bias}} + \underbrace{\|u_{h,k}^\varepsilon - \tilde{u}_{h,k}^{\varepsilon, \tau}\|}_{\text{algebraic error}}.$$

A more detailed target is $E_{tot} \leq C(\eta_h + \eta_k + \eta_\varepsilon + \eta_{alg})$; for viscous Burgers, add $\eta_{nl} + \eta_{stab}$.

5 What is already known, and where novelty must sit (revised)

5.1 Background, not novelty

Continuous null controllability of the 1-D heat equation; uniform/relaxed controllability of semi- and fully-discrete parabolic schemes; penalized HUM as a mature numerical framework; Carleman-weighted primal/dual formulations; strong convergence of primal heat-control approximations; polynomial condition-number growth for particular primal formulations; a priori/a posteriori state-control error estimates in parabolic optimal control generally.

5.2 Novelty that remains plausible, re-ranked after review (iteration 2)

1. **(Strongest)** A rigorous a posteriori decomposition of control error into spatial, temporal, regularization, and algebraic components, specialized to the HUM/Gramian operator rather than a generic tracking cost — the expanded literature appendix below shows this specific specialization is not yet done.
2. **(Strong)** A stopping criterion $\tau(h, k, \varepsilon)$ that prevents algebraic oversolving *in the controllability Gramian*, contrasted against the generic PDE-constrained-optimization stopping rules of Ern-Vohralik and Guttel-Pearson.
3. **(Solid, second-paper scale)** An error/work comparison of HUM versus primal/least-squares formulations at matched terminal and control accuracy.
4. **(Valuable but hard)** Quantitative spectral scaling law for the discrete HUM/penalized-HUM operator under independent and coupled (h, k) refinement, ideally proved rather than only observed.
5. **(Narrower than iteration 1 assumed)** A mesh-dependent regularization/relaxation law $\varepsilon = \varepsilon(h, k)$ balancing bias against discretization and algebraic error. Item A12 in the Priority A appendix below (Boyer & Hernandez-Santamaria, 2026) already proposes choosing a relaxation scale $\phi(\tau) = \tau^p$ according to the accuracy of the numerical method for a time-discrete moment-method controllability result. That is not the same object as the penalized-HUM $\varepsilon(h, k)$ studied here, but it is close enough that this claim can no longer be presented as an unclaimed idea — the paper must state explicitly how its $\varepsilon(h, k)$ rule differs from A12's $\phi(\tau) = \tau^p$ device, or fold the comparison in directly.
6. **(Weakest as a standalone claim)** A preconditioner with mesh-independent iteration counts. This is now one demonstration among many recent ones (see the Priority B appendix below) unless it is shown to exploit structure the generic PDE-constrained preconditioners do not use (e.g., the specific rank structure of the terminal observability Gramian).

6 Standard methodology for a numerical mathematics paper

(Steps 1-8 retained from the original dossier; Step 1's target claim is sharpened below.)

6.1 Step 1: state one primary mathematical claim (sharpened)

For a canonical finite-difference/finite-element HUM discretization of the 1-D heat equation, derive a computable a posteriori decomposition of the control error into spatial, temporal, regularization, and algebraic components, and a stopping rule $\tau(h, k, \varepsilon)$ that attains the discretization-limited accuracy without oversolving, exploiting the specific spectral structure of the discrete observability Gramian.

This replaces the earlier four-part claim (spectrum + stopping rule + regularization balance + preconditioner) as the *headline* sentence; the preconditioner becomes a possible bonus result (labeled C6 later) rather than a load-bearing one.

6.2 Steps 2-8

Verify the PDE solver independently of control via manufactured solutions and EOC; establish the asymptotic refinement regime using at least four to six levels and consecutive-EOC stability checks; separate error sources at every run ($E_h, E_k, E_\varepsilon, E_{alg}, r_T$); stop the iterative solver at the discretization floor ($E_{alg} \lesssim \theta E_{disc}$); test mesh independence of the *algorithm* (iterations, conditioning, CPU, memory — not just accuracy); report work-precision curves rather than work or error alone; and store enough metadata (discretization, solver, regularization, precision, commit, machine) to reproduce every table and figure exactly.

7 Methodological precedents: core table plus full cross-reference appendix

7.1 Core methodological precedents (verified, confirmed as-is)

Verified directly and confirmed on-topic and correctly characterized in the original dossier: Meidner & Vexler (2007, 2008a, 2008b), Ern & Vohralik (2013), Guttel & Pearson (2022), Rees, Dollar & Wathen (2010), Pearson, Stoll & Wathen (2014), Schiela & Ulbrich (2014), Thiele & Wick (2024), Roache (1998), Celik et al. (2008), and critically **Boyer, Hubert & Le Rousseau (2011, *Numerische Mathematik*)**, confirmed as the single closest precedent: it already analyzes semi-discrete-in-space and fully-discrete approximations of null and trajectory-controllability for parabolic equations under an abstract discretization setting, with θ -schemes in time and relaxed uniform observability estimates in both frameworks. Any (h, k) -joint claim in Paper 1 must be stated as a *refinement* of this paper, not as a fresh discovery of joint dependence.

Blank (2026, *Numerical Algebra, Control and Optimization*) was likewise verified directly: for linear parabolic optimal control with implicit time discretization and conforming finite elements, it proves condition numbers for the reduced problem that are bounded independently of the discretization level in arbitrary space dimension, and proposes an all-at-once preconditioner (based on Greif-Schotzau) with discretization-independent eigenvalue bounds, verified numerically. This is a near-direct precedent for contribution claim C6 (preconditioning) and narrows what “new” can mean there.

7.2 Full cross-reference appendix (restored from the original dossier)

Iteration 1 of this revision compressed the original dossier’s most substantial cross-referencing exercise — the seed-based literature expansion — into six bullet points on preconditioning alone. That was an over-correction: most of the original appendix survives review intact and is more useful kept in full, with the 2024-2026 additions folded directly into Priority B rather than kept as a separate list. It is restored below, with annotations added in *italics* marking which contribution claim (C1-C6, defined in “Contribution claims, re-ranked” further on) each item bears on most directly.

How similarity was assessed (unchanged from the original). A paper ranked as especially close if it studies a parabolic control problem (preferably heat); semi-discrete, time-discrete, or fully-discrete controllability/observability; computes controls through HUM, penalized HUM, or an equivalent optimization system; reports or proves convergence in h and/or k ; studies spectral conditioning, mesh-independent convergence, Krylov iterations, or preconditioning; separates discretization, regularization, linearization, or algebraic error; and organizes its numerical section around refinement tables, EOC, parameter sweeps, iteration counts, or work-precision comparisons.

7.2.1 Priority A — direct numerical-controllability papers

- **A1. Boyer, Hubert & Le Rousseau (2011).** *Numerische Mathematik*. The closest direct reference overall — see above. (*C1, C2*)
- **A2. Boyer, Hubert & Le Rousseau (2010a).** *J. Math. Pures Appl.* Discrete Carleman estimates and uniform controllability for semi-discretized parabolic equations; precursor to A1. (*C1*)
- **A3. Boyer, Hubert & Le Rousseau (2010b).** *SIAM J. Control Optim.* Extends discrete Carleman/spectral-inequality ideas to arbitrary dimension; relevant to high-frequency initial-data stress tests. (*C1, robustness slice*)
- **A4. Ervedoza & Valein (2010).** *Revista Matematica Complutense*. Time-discrete abstract parabolic observability and a HUM-type algorithm; direct precedent for separating spatial and temporal numerical effects. (*C1, C2*)
- **A5. Zheng (2008).** *Asymptotic Analysis*. Time-discrete heat-equation controllability; core k -refinement literature, read alongside A4 and A8 before any novelty claim about time discretization. (*C1*)
- **A6. Boyer & Le Rousseau (2014).** *Ann. IHP C*. Semi-discrete Carleman estimates in arbitrary dimension; relaxed controllability for semilinear semi-discrete parabolic equations; bridge toward the Burgers branch. (*relevant to the Burgers pilot, below*)
- **A7. Nguyen (2015).** *DCDS-B*. Uniform semi-discrete controllability for L^q controls; numerical minimization for 1-D heat with Dirichlet boundary control. (*C1, broadens beyond minimal- L^2*)
- **A8. Boyer & Hernandez-Santamaria (2020).** *ESAIM: COCV*. Time-discrete Carleman estimates and relaxed observability/null-controllability, explicitly linking the discrete Carleman parameter to the time step. (*C1, C5*)
- **A9. Lecaros, Morales, Perez & Zamorano (2023).** *J. Differential Equations*. Fully-discrete 1-D heat equation with dynamic boundary conditions, coupling h and Δt to the discrete Carleman parameter; a recent fully-discrete extension against which coupled-refinement experiments (Block C) should be positioned. (*C1*)
- **A10. Chorfi, El Guermai, Maniar & Zouhair (2024).** Penalized HUM plus conjugate gradients for impulse controls with several boundary-condition types; the same HUM-CG

experimental pipeline used here. (*C2, C5*)

- **A11. Bhandari, Dutta & Kumar (2026).** *Math. Control Relat. Fields.* $\phi(\Delta t)$ -null controllability of a time-discrete coupled 1-D parabolic system; convergence of discrete controls to a continuous control. Reinforces that a first paper cannot claim novelty merely from convergence of time-discrete controls. (*background, not novelty*)
- **A12. Boyer & Hernandez-Santamaria (2026).** *Numerische Mathematik.* Moment-method boundary controllability of time-discrete parabolic systems; explicitly proposes choosing a relaxation scale $\phi(\tau) = \tau^p$ according to the accuracy of the numerical method. **This is the item that most narrows contribution claim C5** (the mesh-dependent regularization law): the paper’s $\varepsilon(h, k)$ rule for penalized HUM must be explicitly differentiated from A12’s $\phi(\tau) = \tau^p$ device for a moment-method formulation, not presented as an unclaimed idea. (*C5 — read this one before writing that section*)
- **A13. Carthel, Glowinski & Lions (1994).** *J. Optim. Theory Appl.* Classical numerical heat-controllability reference; historical baseline for the HUM computational strategy. (*background*)
- **A14. Munch (2004).** *C. R. Math.* 1-D wave equation, but methodologically useful: designs an implicit scheme to be uniformly controllable by construction — a cross-equation template for designing around uniform controllability rather than only measuring it. (*C1, design idea*)
- **A15. Infante & Zuazua (1999).** *ESAIM: M2AN.* Hyperbolic, but shows loss of uniform observability from spurious high frequencies and recovery on filtered low-frequency spaces — a strong template for the high-frequency initial-data robustness slice. (*robustness slice*)

7.2.2 Priority B — closest numerical-analysis / optimal-control methodology, including the crowded 2024-2026 preconditioning cluster

- **B1. Blank (2026).** See above. (*C6*)
- **B2. Pearson, Stoll & Wathen (2012).** *SIMAX.* Regularization-robust preconditioners for time-dependent PDE-constrained optimization, tested across mesh, time step, and regularization jointly — almost exactly the three-parameter robustness structure proposed for (h, k, ε) here. (*C6, and a template for Block P’s robustness axis*)
- **B3. Hintermuller, Kopacka & Volkwein (2009).** *ESAIM: COCV.* Proves mesh independence and develops reduced-space/Schur-complement preconditioners; a good model for what “mesh independent” should mean experimentally. (*C6*)
- **B4. Mathew, Sarkis & Schaerer (2010).** *SIAM J. Sci. Comput.* Block/parareal preconditioners for parabolic optimal control, all-at-once formulation, optimal convergence proved and verified numerically. (*C6*)
- **B5. Schoberl & Zulehner (2007).** *SIMAX.* Foundational saddle-point/KKT preconditioning framework; relevant when deciding reduced-operator vs. all-at-once saddle-point formulation for the Gramian. (*C6, design choice*)
- **B6. Pearson & Wathen (2012).** *NLAA.* Schur-complement approximation with eigenvalue bounds explaining mesh-robust performance. (*C6*)
- **B7. Elvetun & Nielsen (2016).** *SIAM J. Sci. Comput.* Preconditioners with MINRES bounds independent of both regularization and mesh size for *local* controls and boundary observations — directly relevant to this dossier’s planned variation of the control region ω . (*C6, robustness slice*)
- **B8. Mardal, Nielsen & Nordaas (2017).** *BIT Numer. Math.* Regularization- and mesh-robust preconditioners under *limited* observation. (*C6*)
- **B9. Guttel & Pearson (2018).** *IMA J. Numer. Anal.* Deferred correction for parabolic

optimal control to raise time accuracy while reusing low-order solvers/preconditioners — relevant if a higher-order temporal scheme is later compared against implicit Euler. (*future work*)

- **B10. Leveque & Pearson (2022).** *NLAA*. Robust all-at-once solver with Crank-Nicolson time discretization, robust in mesh, regularization, and diffusion. (*C6, template for a higher-order comparison*)
- **B11. Chen & Huang (2020).** *Calcolo*. Boundary-value-method time discretization, block triangular preconditioner, Kronecker-product Schur-complement approximation. (*C6*)
- **B12. Yang & Sun (2022).** *Math. Methods Appl. Sci.* Crank-Nicolson finite-difference schemes for parabolic optimal Dirichlet boundary control; proves unconditional stability and second-order convergence, verified numerically — a clean theorem-then-EOC template Paper 1 should imitate structurally. (*methodology template for C1*)
- **B13 (new, found by independent search, not in the original appendix).** Block ω -circulant preconditioners for the all-at-once system, diagonalizable by FFT in a parallel-in-time fashion, with eigenvalues of the preconditioned matrix provably clustered, for first- and second-order time discretizations. (*C6*)
- **B14 (new).** ϵ -circulant diagonalization-based preconditioners for the backward-Euler all-at-once system, explicitly contrasted against Crank-Nicolson circulant/Toeplitz preconditioners (Hon, Fung & Lin, 2025). (*C6*)
- **B15 (new).** An inexact Alternating Direction Method of Multipliers (ADMM) approach for constrained parabolic optimal distributed control. (*C6*)
- **B16 (new).** “Stabilize-then-optimize”: feedback transformations as preconditioners, demonstrated across elliptic, parabolic, and hyperbolic optimal control with explicit condition-number improvement claims. (*C6*)
- **B17 (new).** A spectral-calculus / rational-Krylov approach to optimal control of parabolic equations that is discretization-agnostic by construction. (*C6, alternative paradigm worth citing as a contrast*)

Implication of B13-B17. These did not exist, or were not yet indexed, when the original appendix was assembled, and their absence from a “closest literature” search is itself informative: this is one of the fastest-moving corners of numerical PDE-constrained optimization right now. A bare demonstration of “mesh-independent iteration count” is not, by itself, a publishable increment in 2026 given B1-B17 collectively. If C6 is pursued, the paper must show the proposed preconditioner is either provably/empirically superior to at least two of B1-B17 on the *controllability* problem specifically, or that it exploits a structural feature of the HUM Gramian (e.g., its relation to the terminal observability operator) that none of B1-B17 use. Absent either, C6 belongs inside the C1/C2 paper as a supporting result, not as a headline claim.

7.2.3 Priority C — error decomposition, adaptive stopping, and verification templates

- **C1’.** **Dabaghi, Martin & Vohralik (2020).** *CMAME*. Distinguishes discretization, linearization, and algebraic error in a parabolic problem, stopping nonlinear/algebraic iterations once their contribution no longer materially affects the total error — the strongest concrete template for replacing a fixed solver tolerance with an error-balanced stopping rule. (*C2, primary template*)
- **C2’.** **Becker, Kapp & Rannacher (2000).** *SIAM J. Control Optim.* Foundational goal-oriented a posteriori error control and adaptive finite elements for optimal control. (*C1, methodology template*)

- **C3’**. Liu, Ma, Tang & Yan (2004). *SIAM J. Numer. Anal.* A posteriori estimates for discontinuous-Galerkin time stepping in parabolic optimal control, state and control errors both. (*C1, directly relevant to separating temporal vs. control error*)
- **C4’**. Hinze (2005). *Comput. Optim. Appl.* Variational discretization for control-constrained optimization with optimal control-error estimates — a benchmark for what a mathematically meaningful *control*-discretization error result looks like, as opposed to only a state-PDE error result. (*C1*)
- **C5’**. Becker, Meidner & Vexler (2007). *Optim. Methods Softw.* Space-time finite elements plus second-order optimization algorithms plus large-scale implementation, treated as one computational pipeline. (*methodology template*)
- **C6’**. Gaspoz, Kreuzer, Veese & Wollner (2025). *Math. Control Relat. Fields.* Modern a posteriori analysis for PDE-constrained optimization including the effect of *small regularization parameters* — directly relevant to the regularization error branch (C5). (*C1, C5*)
- **C7’**. Zhang, Zhao & Hou (2025). *Appl. Numer. Math.* A priori state/control error estimates for a space-time discretization of a parabolic optimal-control problem with numerical rate verification; close in reporting structure even though the model includes a time delay. (*methodology template*)

7.2.4 What the full appendix changes, net of iteration 1 and iteration 2

The expanded and now-restored literature map sharpens the novelty boundary further than iteration 1 alone did. A publishable Paper 1 should not claim, in isolation: convergence of discrete controls as h or $k \rightarrow 0$ (A1, A4, A5, A11); relaxed controllability under mesh/time coupling (A2, A3, A6, A8, A9); a generic demonstration that preconditioning reduces iterations (B1-B17, now a saturated claim); or a mesh-dependent relaxation/regularization scale in the abstract (A12). What survives as plausibly open, after the full pass, is the combination: a computable a posteriori decomposition (C1) *paired with* a provable stopping rule *derived from it* (C2), specialized to the HUM/Gramian operator rather than a generic tracking-type cost — because no single item in Priority A, B, or C does both of those things together for this specific operator.

8 Exact experiment protocol and experiment count (revised)

8.1 Step 0 (new): pilot phase before committing the full budget

Before running Blocks C-P at full scale, run a **pilot slice**: Block V in full (15 configurations, unchanged), plus a **reduced** Block C using only 3 of the 6 planned mesh levels on the coupled path $k = Ch$. Purpose: confirm, cheaply, that

1. an asymptotic (not pre-asymptotic, not roundoff-dominated) regime is reached within the planned mesh range at all;
2. the proposed acceptance thresholds further on (0.1x, 5%, 20%, effectivity band) are achievable at the planned precision, or need to be loosened/tightened;
3. the condition-number growth is smooth enough that a slope claim (see the thresholds table, “condition-number power law”) is meaningful over the planned window.

Gate 0. Do not proceed to the full 123-configuration core until the pilot shows a stable asymptotic window of at least three consecutive mesh levels. If it does not, widen the mesh range or revisit the discretization before spending the rest of the budget. This gate did not exist in the original protocol and is the main structural fix from this review.

8.2 Blocks V, C, S, R, P (retained from the original dossier)

- **Block V — verification, 15 configurations.** Spatial order (6, $N \in \{20, 40, 80, 160, 320, 640\}$), temporal order (6, $M \in \{50, 100, 200, 400, 800, 1600\}$), reference-control consistency (3, nested high-resolution meshes).
- **Block C — conditioning and refinement, 24 configurations.** Spatial sweep (6), temporal sweep (6), coupled path $k = Ch$ (6), coupled path $k = Ch^2$ (6). Record $\lambda_{\min}, \lambda_{\max}, \kappa, E_u, r_T, \|u\|$, Krylov iterations, work.
- **Block S — solver tolerance and oversolving, 30 configurations.** Six production meshes $x \tau \in \{10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}\}$. Primary plot: $E_u(\tau), E_{alg}(\tau), N_{it}(\tau)$, identifying the plateau.
- **Block R — regularization path, 30 configurations.** Six meshes $x \varepsilon \in \{0, h^2, h^4, h^6, h^8\}$. **Pre-registration added:** before running this block, record a predicted qualitative shape for the bias-vs-conditioning Pareto curve (e.g., “bias should fall roughly like ε while κ should grow roughly like ε^{-1} near the small- ε end”) so the block tests a hypothesis rather than only producing a curve to fit after the fact.
- **Block P — preconditioning, 24 configurations.** Four solver/preconditioning choices (none, diagonal/scaled, incomplete factorization, operator/block candidate) x six meshes. Given the crowded 2024-2026 baseline cluster above, add a fifth column comparing against at least one published 2024-2026 preconditioner from that list as an external baseline, not only the internal “none” baseline.

8.3 Minimum core and robustness count (unchanged totals)

$$15 + 24 + 30 + 30 + 24 = 123 \text{ primary configurations.}$$

Recommended robustness slices: $T \in \{1, 0.5, 0.25, 0.125\}$ on 3 meshes (12); $|\omega| \in \{0.8, 0.5, 0.25, 0.1\}$ on 3 meshes (12); $m \in \{1, 2, 4, 8, 16, 32\}$ for $y_0 = \sin(m\pi x)$ on 3 meshes (18). **Pre-registration added:** state in advance what a robustness-slice failure would mean for the headline claim (e.g., “if condition number depends materially on $|\omega|$, the C3 scaling law must be restated as ω -dependent,

not universal”) so these slices can falsify, not just decorate, the main result. Total recommended study: $123 + 12 + 12 + 18 = 165$ configurations, run only after Gate 0 and Gate 1 (see the revised plan below) are passed.

8.4 Timing and exclusions (unchanged)

Deterministic numerical results are not repeated; timing configurations are repeated five times after a warm-up run and summarized by median/IQR. Burgers, age-dependent systems, all control regions, and all least-squares variants remain excluded from Paper 1.

9 Acceptance criteria and numerical thresholds (now provisional)

The thresholds below are retained from the original dossier but are now explicitly labeled **provisional, pending pilot confirmation (Step 0)**. None should be treated as fixed until the pilot phase shows they are attainable at the planned mesh/precision budget.

Quantity	Principle	Provisional operational criterion
Forward/adjoint EOC	Observed order approaches theory in the asymptotic regime	Accept only when at least 3 consecutive EOCs are stable and the mean is within approximately 15% (or 0.2 absolute order) of theory; report failures rather than tuning them away
Reference-control accuracy	Reference must be substantially more accurate than the candidate	Difference between two finest references $\leq 0.1 \times$ candidate error, else refine further
Algebraic stopping	Iterative error should not materially affect total error	$E_{alg} \leq 0.1 E_{disc}$ when an estimator exists; otherwise a plateau rule: largest τ giving at most 5% degradation vs. tightest solve
Terminal null residual	Application/formulation-dependent, no universal value	Report normalized $r_T/\ y_0\ $ at three bands (10^{-4} , 10^{-6} , 10^{-8})
Condition-number power law	Meaningful only in a stable asymptotic range	≥ 4 asymptotic points; slope change less than 10% between nested fit windows; report CI and residuals
Preconditioner robustness	“Mesh independent” = iteration count does not grow materially with dimension	Variation at most 20% across the production sequence, or fitted exponent statistically indistinguishable from zero — and comparison against an external 2024-2026 baseline (Priority B appendix)
Estimator effectivity	Reliable if bounded; sharp if near 1	Exploratory: $[0.5, 2]$ and stabilizing; sharpness claim: asymptotic $[0.8, 1.25]$
Roundoff onset	Refinement should stop when discretization gain is masked by floating-point error	Flag first level where error fails to drop at least 20% despite refinement and a tighter solver tolerance; confirm in higher precision
Timing	Runtime is noisy; accuracy is deterministic	Five repeats, median/IQR, fixed hardware/software state

Why still provisional. The original dossier asserted these constants without benchmarking them against what the cited precedent papers actually achieved numerically. The pilot phase (Step 0) exists specifically to check, on a small slice, whether e.g. a 0.1x algebraic-to-discretization ratio and a 15% EOC tolerance are simultaneously achievable at double precision on the planned mesh range, or whether one of them needs to be relaxed before the full 165-run budget is spent chasing an unattainable target.

10 Primary paper: a publishable scope (revised)

10.1 Provisional title

Solver-aware error decomposition for numerical HUM control of the one-dimensional heat equation: a computable stopping rule under space-time refinement.

(Changed from the original title to foreground C1/C2 — the strongest claims after review — rather than leading with conditioning/regularization/preconditioning, which foregrounded the weakest one.)

10.2 Contribution claims, re-ranked (C1-C6, defining the labels used throughout)

- **C1 (headline).** A computable a posteriori decomposition of control error into spatial, temporal, regularization, and algebraic components, specialized to the discrete HUM/Gramian operator.
- **C2 (headline, paired with C1).** A stopping rule $\tau(h, k, \varepsilon, \eta_{target})$ that provably or demonstrably prevents algebraic error from dominating the target accuracy, contrasted explicitly with the generic rules of Ern-Vohralik and Guttel-Pearson (Priority C appendix item C1').
- **C3 (valuable but hard).** A quantitative spectral scaling law for the discrete HUM/penalized-HUM operator under independent and coupled (h, k) refinement, ideally proved rather than only observed, positioned as a refinement of Boyer, Hubert & Le Rousseau (A1) rather than a fresh discovery.
- **C4 (solid, second-paper scale).** An error/work comparison of HUM versus primal/least-squares formulations at matched terminal and control accuracy — scoped as Paper 2, below.
- **C5 (narrowed, needs explicit differentiation).** A mesh-dependent regularization law $\varepsilon(h, k)$ balancing terminal error, bias, and conditioning, tested against a pre-registered Pareto-shape hypothesis in Block R, and stated explicitly against A12's $\phi(\tau) = \tau^p$ device (Priority A appendix) so it is not presented as an unclaimed idea.
- **C6 (bonus, conditional).** A preconditioner or scaling result, reported only if it beats or is mechanistically distinct from the B1-B17 baselines in the Priority B appendix above; otherwise omitted or moved to a “preliminary observations” subsection.

A paper does not need all six. C1+C2 together, specialized to the Gramian, is enough for a strong numerical-analysis submission on its own; C3 or C5 as a secondary result strengthens it further; C6 should be attempted only if time remains after C1/C2 are solid, given how contested that specific claim now is.

10.3 Minimum theorem target (derived, not placeholder)

For the discrete SPD operator $A_{h,k}^\varepsilon$ with exact solution $p_{h,k}^\varepsilon$ and inexact CG iterate $\tilde{p}_{h,k}^{\varepsilon,\tau}$ at residual $r_{alg} = b - A_{h,k}^\varepsilon \tilde{p}_{h,k}^{\varepsilon,\tau}$, the elementary bound

$$\|p_{h,k}^\varepsilon - \tilde{p}_{h,k}^{\varepsilon,\tau}\|_2 \leq \frac{\|r_{alg}\|_2}{\lambda_{\min}(A_{h,k}^\varepsilon)}$$

already links algebraic residual to spectral conditioning. The concrete theorem target is to (i) obtain a computable, run-time-available **lower bound estimate** $\underline{\lambda}_{\min}(A_{h,k}^\varepsilon) \leq \lambda_{\min}(A_{h,k}^\varepsilon)$ from the CG Lanczos recurrence already being computed during the solve (at no extra matrix-vector products),

- (ii) combine it with an a priori or a posteriori discretization estimate $\eta_{disc}(h, k, \varepsilon)$ from Step 4, and
- (iii) derive the **implementable** stopping rule

$$\|r_{alg}\|_2 \leq \theta \lambda_{\min}(A_{h,k}^\varepsilon) \eta_{disc}(h, k, \varepsilon), \quad 0 < \theta < 1,$$

then prove that this rule guarantees $E_{alg} \leq \theta E_{disc}$ up to the approximation error in $\lambda_{\min}$ and η_{disc} themselves. This is a modest but genuine theorem: it upgrades the “elementary bound” already present in the original dossier from an observation into an actionable, provable stopping criterion, which is exactly what distinguishes C2 from a re-run of Ern-Vohralik in a new setting.

11 Matched HUM vs. primal/least-squares study: Paper 2

(Unchanged.) Two formulations x six mesh levels x three accuracy targets = 36 matched configurations, comparing $E_u, r_T, \|u\|, \kappa, N_{it}$, CPU, and memory at matched target accuracy, with the goal of identifying spectral equivalence after scaling or a proof that one formulation has worse (h, k, ε) -dependence.

12 Burgers as Paper 3, and the age-dependent transfer

(Unchanged in substance.) A deliberately small pilot ($A \in \{0.01, 0.1, 0.5, 1\}$, $\nu \in \{10^{-1}, 10^{-2}, 10^{-3}\}$, $N \in \{80, 160\}$, 24 configurations) to identify when the linear-heat error budget stops explaining the observed error, run only after the linear study is stable. Age-dependent/structured population systems remain a longer-term transfer target, not a first benchmark.

13 Revised plan (updated for the pilot gate)

Month 1. Week 1: solver architecture, reproduce thesis results. Week 2: Block V. Week 3: reference hierarchy, spectral diagnostics. **Week 4: pilot phase (Step 0) — reduced Block C, check Gate 0.**

Gate 0 (new). Do not proceed to the full Block C-P budget unless the pilot shows a stable asymptotic window and the thresholds table further on are attainable as stated (or have been revised in response to the pilot).

Month 2. Full Blocks C, S; attempt the algebraic-error certificate from the theorem target above; full Block R against the pre-registered Pareto hypothesis; formulate mesh-dependent τ, ε hypotheses.

Gate 1 (unchanged). A tolerance rule is credible only if it predicts the plateau across several mesh levels without being retuned per grid.

Month 3. Block P, including the external-baseline comparison (Priority B appendix); robustness slices with pre-registered failure criteria; work-precision diagrams; internal draft with only verified claims.

Gate 3 (unchanged). If no preconditioner beats or differs mechanistically from the 2024-2026 baselines, the fallback contribution is the quantified C1/C2 mechanism and stopping result — not an unsupported robustness claim.

Month 4. Validate one primal/least-squares implementation; run a subset of the 36 matched configurations; Burgers pilot only if the heat analysis is stable; freeze plots/tables; choose scope and venue class using the decision tree below.

14 Publication-readiness decision tree (unchanged)

1. Stable scaling + theorem/bound + useful stopping rule -> write Paper 1 immediately.
2. Stable scaling + preconditioner distinct from the Priority B baselines + mesh-independent iterations -> target a scientific-computing / numerical-linear-algebra venue.
3. Reliable a posteriori error decomposition alone -> target a strong numerical-analysis paper, potentially with adaptivity.
4. Only empirical curves, no new mechanism -> do not publish yet; use the data to design the theorem/preconditioner.
5. HUM and least-squares show a robust, unexplained spectral gap -> make the formulation comparison the primary paper.
6. No clean single exponent -> identify multiple regimes (pre-asymptotic, asymptotic, roundoff) and test whether regime boundaries are predictable from nondimensional quantities.

15 Recommended manuscript structure (unchanged)

Introduction and novelty (stated explicitly against Boyer, Fernandez-Cara-Munch, Labbe-Trelat, Gonzalez Casanova-Hernandez-Santamaria, and the 2024-2026 preconditioning cluster in the Priority B appendix); continuous problem and control formulation; discretization; error and conditioning analysis (the theorem); solver and stopping strategy; verification experiments; primary experiments (Blocks C, S, R, P); robustness and work-precision; limitations (explicitly 1-D, discretization-specific, formulation-specific); reproducibility (parameter table, code/version, run IDs).

16 Experiment-tracking schema (unchanged)

Every run stores: run id, paper block, PDE (heat/Burgers), formulation (HUM/penalized-HUM/primal-LS), ν, T, ω , initial condition, space/time discretization, regularization rule, solver/tolerance/preconditioner, reference run id and error estimate, and outputs (control error, terminal residual, control norm, $\lambda_{\min}, \lambda_{\max}, \kappa$, iterations, CPU, memory, EOC, effectivity), plus provenance (git commit, machine, precision, date).

17 Final assessment (revised)

The project is a publishable *direction*, not yet a publishable *result*. The independent review confirms the core precedent set is real and correctly used, but narrows where genuine novelty can sit: the a posteriori error decomposition and solver-aware stopping rule specific to the controllability Gramian (C1/C2) are the strongest candidates; the mesh-dependent regularization law (C5) is a solid second leg, though it must be explicitly differentiated from A12; the preconditioning claim (C6) is the weakest as a standalone contribution given how active that specific sub-field became in 2024-2026, and should be pursued only as a conditional bonus benchmarked against named external baselines. The revised protocol adds a pilot gate before the full 165-configuration budget is spent, and replaces the placeholder theorem with an actual, checkable derivation target. The most defensible first-paper objective is now:

For a canonical 1-D heat null-control discretization, derive a computable a posteriori decomposition of control error into discretization, regularization, and algebraic components, and a provable stopping rule that reaches discretization-limited accuracy without unnecessary computation — verified, not merely illustrated, by the staged experiment protocol above.

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